Materials Properties MCQs - Exam Style

Q1) Which of the following statements about non-linear elastic materials is TRUE?

A) They exhibit permanent deformation after unloading.

B) The stress-strain curve is linear at all strain levels.

C) The material fully recovers its original shape after unloading.

D) The elastic modulus is constant throughout deformation.

Q2) Which material property is most directly linked to a material's ability to resist the propagation of a pre-existing crack?

A) Toughness

B) Yield Strength

C) Hardness

D) Fracture Toughness

Q3) Which of the following correctly describes the Poisson's ratio (nu) of a material?

A) It is the ratio of axial strain to lateral strain under axial loading.

B) It is always greater than 1 for elastic materials.

C) It is the negative of the ratio of lateral strain to axial strain.

D) It describes the hydrostatic stress response of a material.

Q4) Which of the following correctly explains why the modulus of rupture is often higher than tensile strength in ceramics?

A) Bending loads introduce compressive stresses that strengthen the material.

B) Ceramics become tougher under bending due to work hardening.

C) In bending, only a small volume experiences maximum stress, reducing the chance of encountering a flaw.

D) The surface energy of ceramics increases under tensile loading, leading to early failure.

Q5) Which condition would most likely increase the electrical resistivity of pure copper?

A) Annealing the copper to reduce dislocations

B) Increasing the copper's grain size

C) Cold-working the copper to introduce more defects

D) Cooling the copper to cryogenic temperatures

Q6) Which of the following is the most appropriate method for accurately determining the Young's modulus of a material?

A) Measuring permanent elongation under tensile loading

B) Calculating surface hardness with a Brinell test

C) Measuring natural vibration frequency of a beam

D) Observing plastic deformation during compression

Q7) A rubber material is stretched to 500% strain and then unloaded. Which of the following best describes its mechanical behaviour?

A) It undergoes plastic deformation with necking.

B) It returns to its original length, showing linear elastic behaviour.

C) It returns to its original shape, showing non-linear elastic behaviour.

D) It shows permanent set due to anelastic energy loss.

Q8) Why does proof stress (e.g. 0.1% proof) need to be defined for some materials?

A) Because their elastic limit is too high to measure directly

B) Because they fracture before yielding

C) Because they do not exhibit a well-defined yield point

D) Because their stress-strain curve is perfectly linear

Q9) Why does thermal conductivity decrease when zinc is added to copper?

A) Zinc increases the density of the alloy, reducing electron mobility.

B) Zinc atoms scatter heat-carrying electrons, disrupting thermal transport.

C) Zinc changes the crystal structure of copper to an amorphous state.

D) Zinc increases the heat capacity of copper, reducing conductivity.

Q10) Which of the following would most likely result in a material appearing opaque rather than transparent?

A) Low porosity and single crystal structure

B) High porosity in a polycrystalline structure

C) Low thermal expansion coefficient

D) High magnetic permeability

Q11) Which of the following statements about hardness testing is most accurate?

A) A harder material will allow the indenter to penetrate deeper.

B) The Brinell test is preferred for measuring fracture toughness.

C) The depth of penetration is inversely related to the material's yield strength.

D) Hardness is measured using thermal expansion under load.

Answer Key

Q1) C

Q2) D

Q3) C

Q4) C

Q5) C

Q6) C

Q7) C

Q8) C

Q9) B

Q10) B

Q11) C